



PRELIMINARY EXAMINATIONS
HIGHER 1

CANDIDATE
NAME

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CIVICS
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CHEMISTRY

8873/02

Paper 2 Structured Questions

23 August 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Civics Group, centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided

The number of marks is given in brackets [] at the end of each question or part question.

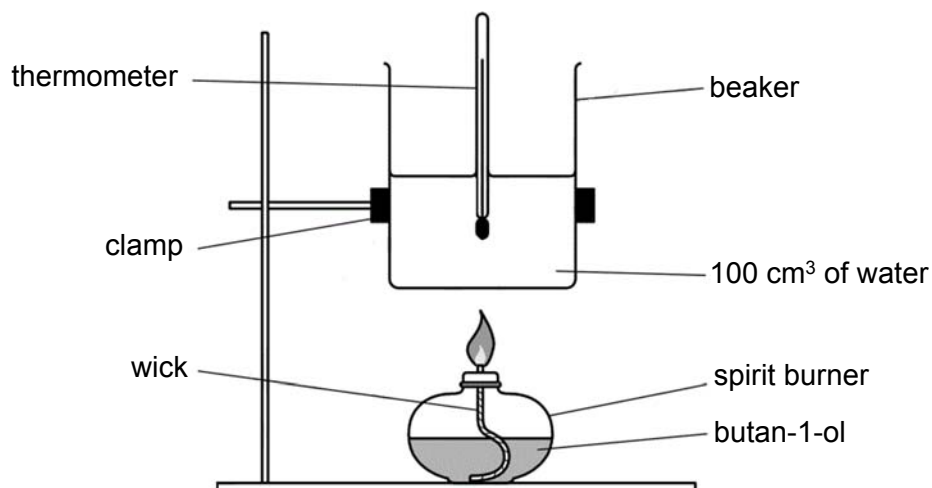
For Examiner's Use	
Section A	/ 60
1	/ 16
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Total	

This document consists of **22** printed pages.

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 (a) For many compounds, the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy change of combustion can be used. The enthalpy change of combustion of butan-1-ol can be found by a calorimetry experiment in which the heat given off during combustion is used to heat a known mass of water and the temperature change recorded.



temperature of water before heating = 25.0 °C

temperature of water after heating = 66.1 °C

mass of spirit burner and butan-1-ol before heating = 80.44 g

mass of spirit burner and butan-1-ol after heating = 79.70 g

- (i) The heat transfer from the spirit burner to water is known to be only 70% efficient. From the data provided above, determine the enthalpy change of combustion of butan-1-ol.

$$\text{mass of butan-1-ol reacted} = 80.44 - 79.70 = 0.74 \text{ g}$$

$$\text{No. of moles of butan-1-ol reacted} = 0.74 / 74 = 0.0100 \text{ mol}$$

$$\text{amount of heat taken in by water} = mc\Delta T$$

$$= 100 \times 4.18 \times (66.1 - 25.0)$$

$$= 17.2 \text{ kJ}$$

$$\bullet \text{ Heat provided by butan-1-ol} = 100 \times 17.2 / 70 = 24.6 \text{ kJ}$$

$$\text{enthalpy change of combustion of butan-1-ol} = - 24.6 / 0.0100$$

$$\bullet = - 2460 \text{ kJ mol}^{-1}$$

[3]

- (ii) Write the chemical equation that represents the standard enthalpy change of combustion of butan-1-ol, including the state symbols.



[1]

- (iii) Use the bond energies given in the Data Booklet to calculate another value for the standard enthalpy change of combustion of butan-1-ol.

Bonds broken: 3 C-C bonds, 9 C-H bonds, 1 C-O bond, 1 O-H bond
6 O=O bonds

Bonds formed: 8 C=O bonds, 10 O-H bonds

- $\left[\begin{array}{l} \text{Energy required to break bonds} = (3 \times 350) + (9 \times 410) + (1 \times 360) + (1 \times 460) + (6 \times 496) = 8536 \text{ kJ} \\ \text{Energy released from bonds formed} = (8 \times 805) + 10 (460) = 11\,040 \text{ kJ} \end{array} \right.$
- $\Delta H_c^\ominus$ of butan-1-ol = $-11\,040 + 8536 = -2504 \text{ kJ mol}^{-1}$

[2]

- (iv) Suggest a reason for the discrepancy between this value and that calculated in (a)(i).

- The bond energies given in the data booklet are only averages and would not apply to the exact compounds in the reaction.

Or

The $\Delta H_c^\ominus$ calculated applies for the reactants and products in the gaseous phase but the standard enthalpy change of combustion is defined in terms of the formation of liquid H_2O rather than gaseous H_2O .

Reject: heat loss to surrounding in (a)(ii), as that is accounted by the 70% efficiency

[1]

- (b) The table below gives some data on four fuel sources: methanol, ethanol, hydrogen and octane. Octane can serve as a rough approximation of petrol.

name	formula	molar mass/ g mol^{-1}	density/ g cm^{-3}	$\Delta H_c^\ominus$ (298K)/ kJ mol^{-1}
methanol	CH_3OH	32	0.793 ^a	-726.0
ethanol	$\text{CH}_3\text{CH}_2\text{OH}$	46	0.789 ^a	-1367.3
liquid hydrogen	H_2	2	0.0711 ^b	-285.8 [#]
octane	C_8H_{18}	114	0.703 ^a	-5470.2

^a At 298K and 1 bar pressure

^b At 20K and 1 bar pressure

standard enthalpy change of combustion of hydrogen **gas**

- (i) Calculate the density of gaseous hydrogen at 298 K and 1 bar pressure. Assume 1 mol of any gas occupies 24.7 dm³ at 298 K and 1 bar pressure. Give your answers in g cm⁻³.

Molar mass of H₂ = 2.0 g

Volume of 1 mol of H₂ = 24.7 dm³ = 24 700 cm³

Density of gaseous hydrogen = mass/ volume

$$= 2.0/ 24700$$

$$\bullet \quad = 8.10 \times 10^{-5} \text{ g cm}^{-3}$$

[1]

- (ii) An important property of a fuel, especially when the fuel has to be lifted such as in aviation, is the energy released on combustion per gram of fuel.

Calculate the enthalpy change of combustion per gram of fuel at 1 bar pressure and 298K for methanol.

$$-726 / 32 = -22.7 \text{ kJ g}^{-1}$$

[1]

- (iii) Another important characteristic of a fuel, especially when there is a fuel tank of limited size, is the energy released on combustion per cm³ of fuel.

Calculate the enthalpy change of combustion per cm³ of fuel for octane.

$$-5470.2/ (114/0.703) = -33.7 \text{ kJ cm}^{-3}$$

[1]

- (iv) Explain why, given the data in the question, it is not strictly possible to make a fair comparison of the energy released per cm³ of liquid hydrogen with other fuels.

vaporisation of hydrogen not accounted for

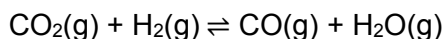
OR

the value of the density of liquid hydrogen is unsuitable for the calculation of energy per unit volume for gaseous hydrogen

Or enthalpy change of combustion value for hydrogen is for standard conditions, and so relates to **gaseous hydrogen**, not to liquid hydrogen.

[1]

- (c) Carbon monoxide and steam are products of incomplete combustion of hydrocarbons. An equilibrium mixture of carbon monoxide and steam is also produced when carbon dioxide is mixed with hydrogen gas.



- (i) 0.0100 mol of CO_2 and 0.0200 mol of H_2 were mixed and allowed to reach equilibrium at a temperature T. The final mixture contained 6.93×10^{-3} mol of H_2O .

Calculate the value of K_c at this temperature

	$\text{CO}_2(\text{g}) + \text{H}_2(\text{g})$		$\rightleftharpoons$	$\text{CO}(\text{g})$	$+ \text{H}_2\text{O}(\text{g})$
initial amount/ mol	0.0100	0.0200		0	0
change in amount/ mol	-6.93×10^{-3}	-6.93×10^{-3}		$+6.93 \times 10^{-3}$	$+6.93 \times 10^{-3}$
eqm amount/ mol	1.31×10^{-2}	3.07×10^{-3}		6.93×10^{-3}	6.93×10^{-3}

- working

$$K_c = \frac{\left(\frac{6.93 \times 10^{-3}}{V}\right)\left(\frac{6.93 \times 10^{-3}}{V}\right)}{\left(\frac{1.31 \times 10^{-2}}{V}\right)\left(\frac{3.07 \times 10^{-3}}{V}\right)}$$

- = 1.20

[2]

- (ii) The numerical value of equilibrium constant is 0.08 at 400°C and 0.40 at 600°C . Explain whether the reaction between carbon dioxide and hydrogen is exothermic or endothermic.

- The position of equilibrium shifts right when temperature increases.
- By Le Chatelier's Principle, this indicates that forward reaction absorbs heat to counteract the change exerted on the system, hence the reaction is endothermic.

[2]

- (iii) State and explain the effect of increasing pressure on the position of this equilibrium.

- The position of equilibrium does not change since there are equal number of moles of gases on each side of the equation.

[1]

[Total: 16]

- 2 (a) What do you understand by the *Brønsted–Lowry* theory of acids and bases.

Bronsted-Lowry acids are proton donors and bases are proton acceptors.

[1]

- (b) The following is a list of compounds that react with or dissolve in water:

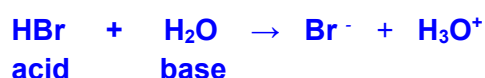
Sodium chloride, hydrogen bromide, silicon tetrachloride, ammonia, methanol.

Water can react as either an acid or a base. Choose a compound from the above list with which water acts as

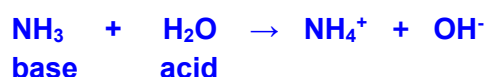
- a *Brønsted* base
- a *Brønsted* acid

Construct a balanced equation for each reaction.

- water as a *Brønsted* base



- water as a *Brønsted* acid



[2]

- (c) Lactic acid is a monoprotic acid which is an important flavouring component of many foods such as cheese, yoghurt and pickled cabbage. A solution of lactic acid in water containing 0.10 mol dm^{-3} has a pH of 2.43.

- (i) Is lactic acid a strong or weak acid? Explain your answer.

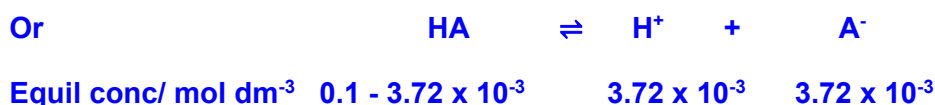
- **Weak acid.**

$[\text{H}^+] = 10^{-2.43} = 3.72 \times 10^{-3} \text{ mol dm}^{-3} < 0.1 \text{ mol dm}^{-3}$. Hence, lactic acid is partially dissociated.

[1]

- (ii) Use the data given to calculate the value of K_a for lactic acid.

$$K_a = [\text{H}^+]^2 / [\text{HA}]_{\text{initial}} = (3.72 \times 10^{-3})^2 / (0.1) = 1.38 \times 10^{-4} \text{ mol dm}^{-3}$$



$$K_a = [\text{H}^+]^2 / [\text{HA}]_{\text{equilibrium}} = 1.43 \times 10^{-4} \text{ mol dm}^{-3}$$

[2]

- (iii) Suggest a suitable indicator for the titration of lactic acid with aqueous barium hydroxide.

- **Phenolphthalein or thymol blue**

[1]

- (iv) A sample of lactic acid was extracted from a blood sample. It was dissolved in water and titrated with $0.250 \text{ mol dm}^{-3}$ barium hydroxide. It was found that 22.3 cm^3 of hydroxide was required for neutralisation. Calculate the mass of lactic acid in the blood sample. (M_r of lactic acid = 90.0)

Amount of $\text{Ba}(\text{OH})_2$ reacted = $0.250 \times 22.3/1000 = 5.575 \times 10^{-3} \text{ mol}$

• Amount of lactic acid = $2 \times 5.575 \times 10^{-3} = 0.0112 \text{ mol}$

• mass of lactic acid = $0.0112 \times 90.0 = 1.008$ or 1.01 g

[2]

- (v) The mass composition of carbon, hydrogen and oxygen in lactic acid is 40.0%, 6.7% and 53.3% respectively. Determine the molecular formula of lactic acid.

	C	H	O
Mass /g	40.0	6.7	53.3
No. of moles	$40.0/12 = 3.3$	6.7	$53.3/16 = 3.3$
Ratio	1	2	1

Empirical formula is CH_2O

$n[12.0 + 2(1.0) + 16.0] = 90.0$

$n = 3$

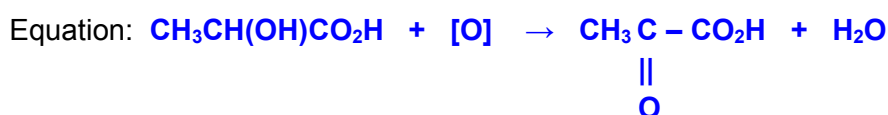
Molecular formula is $\text{C}_3\text{H}_6\text{O}_3$

[1]

- (vi) When lactic acid is strongly oxidised, it gives an organic product with the same number of oxygen atoms. State the reagent and condition for the strong oxidation to occur.

Write a balanced equation for the oxidation reaction showing clearly the structural formula of lactic acid and its oxidation product.

Reagent and condition: **acidified $\text{K}_2\text{Cr}_2\text{O}_7$, heat Or acidified KMnO_4 , heat**

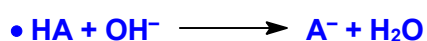


[2]

- (d) Explain, using an equation why an aqueous mixture of lactic acid and sodium lactate can act as a buffer solution on addition of alkali.

You may represent lactic acid as HA and sodium lactate as Na^+A^- .

On addition of an alkali, HA will react with added OH^- resulting in no significant change in pH.



[1]

[Total: 13]

3 (a) Outline the principles of the Valence Shell Electron Pair Repulsion (VSEPR) theory.

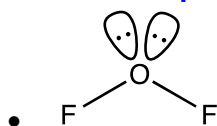
- **Electron pairs (lone and bond pairs) in the outer shell of the central atom will arrange themselves as far apart as possible so as to minimise repulsion and maximise stability.**
- **The lone pair- lone pair repulsion is greater than lone pair-bond pair repulsion which in turn is greater than bond pair-bond pair repulsion.**

[2]

(b) Predict and draw the shapes of the following molecules or ions.

(i) F_2O

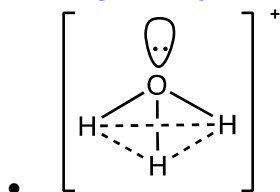
- **Bent shape**



[1]

(ii) H_3O^+

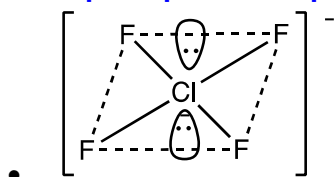
- **Trigonal pyramidal shape**



[1]

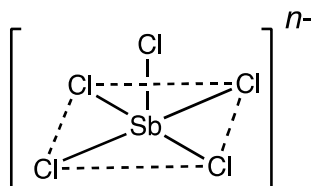
(iii) ClF_4^-

- **Square planar shape**



[1]

- (c) Antimony, Sb, is in Group 15 of the Periodic Table. It forms a series of salts which contain the SbCl_5^{n-} anion, the structure of which is square pyramidal.



Deduce the total number of electrons around the antimony atom, the value of n and the oxidation number of Sb in this ion.

- **Total no of electrons around Sb atom = 10 + 2 = 12 (there is a lone pair on Sb since it is square-pyramidal)**

Since Sb is in Group 15 it has acquire an additional 2 electrons.

- **Hence $n = 2$ and oxidation number of Sb is +3**

[2]

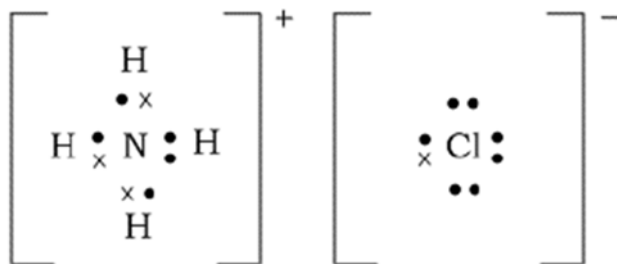
[Total: 7]

- 4 (a) Ammonium chloride, NH_4Cl , is a salt which has covalent bonding, dative bonding and ionic bonding. It is very soluble in water and can be used to maintain the urine at an acidic pH in the treatment of some urinary-tract disorders.

- (i) What is a dative bond? [1]

Dative / Co-ordinate bond is a covalent bond in which a pair of electrons is shared between 2 atoms but ONLY ONE of them provides both electrons that make up the bond.

- (ii) Draw a “dot-and-cross” diagram to show the bonding in NH_4Cl . [1]



(iii) In terms of structure and bonding, explain why NH_4Cl is highly soluble in water.

[2]

- NH_4Cl has a giant ionic structure and water has a simple molecular structure.
- Ion-dipole interactions formed between the ions present in NH_4Cl and polar water molecules releases sufficient energy to overcome the ionic bonds in NH_4Cl and hydrogen bonds between water molecules.

(b) (i) Research in nanotechnology is growing and many new nanomaterials are being produced with ever more uses.

Suggest what is meant by the term nanomaterials.

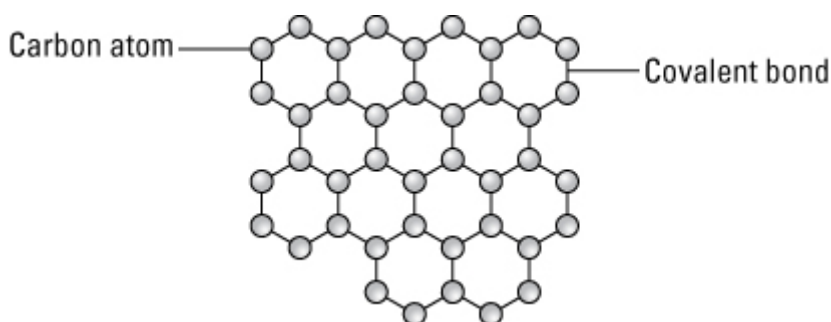
[1]

A nanomaterial is a material with at least one dimension on the nanoscale (length $\leq 100\text{nm}$).

(ii) Describe the structure of graphene. Explain how this structure gives graphene the properties of high strength and high electrical conductivity.

You may find it useful to include a labelled diagram in your answer.

[3]



- The structure of graphene is a giant covalent structure and it is a single layer of hexagons (graphite) where every atom is on the surface of the structure.
- Each carbon atom has strong covalent bonds with three other carbon atoms and a lot of energy is needed to break the bonds hence it has high tensile strength.
- One p electron per carbon atom is delocalised over the graphene layer and is able to act as a mobile charge carrier. Hence it has high electrical conductivity.

[Total: 8]

5 Period 3 elements react with oxygen to form oxides.

(a) Write equations for the reactions, if any, of the following three oxides with water. In each case, describe the effect of the resulting solutions on Universal Indicator solution.

(i) Na_2O [1]

- $\text{Na}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)}$;
The Universal Indicator solution will turn violet or indigo in the solutions.

(ii) Al_2O_3 [1]

- No reaction
The Universal Indicator solution will turn green in the solutions.

(iii) P_4O_6 [1]

- $\text{P}_4\text{O}_6\text{(s)} + 6\text{H}_2\text{O(l)} \rightarrow 4\text{H}_3\text{PO}_3\text{(aq)}$;
The Universal Indicator solution will turn red in the solutions.

(b) The oxides MgO , Al_2O_3 and SiO_2 are all used as refractory materials due to their high melting points. The last two are major constituents of gemstones, such as rubies, sapphires and amethysts.

If a sample of one of the oxides was provided as a white powder, describe the reactions you could carry out on the powder to determine which of the three oxides it was.

[3]

- I will add HCl to a small amount of the powder. If it is soluble in the acid, it could be either MgO or Al_2O_3 .
- Then I will add NaOH to another small portion of the powder. If it is soluble, then it is Al_2O_3 , if not, then it is MgO .
- If the powder is not soluble in acid or base, it is SiO_2 .

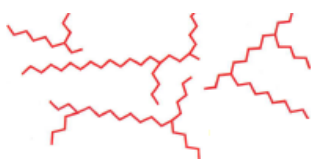
[Total:6]

- 6 (a) Some of the most commonly used polymers are formed by the polymerisation of ethene, C_2H_4 . The presence of side-chains affects the bulk properties of an addition polymer.

Poly(ethene) exists in two different forms LDPE (low density poly(ethene)) which has lots of side-chains, and HDPE (high density poly(ethene)) in which there are fewer and shorter side-chains.

- (i) Explain with the aid of sketches why the presence of side-chains causes a difference in density in poly(ethene).

[2]



LDPE



HDPE

- Minimum of 2 chains suitable sketches
 - LDPE has more space between the chain/polymers, hence, low density or HDPE less empty space and more closely packed between the chains, hence, high density.
- (ii) By reference to the type of bonding between the poly(ethene) chains, explain why LDPE has a lower melting point than HDPE.

[1]

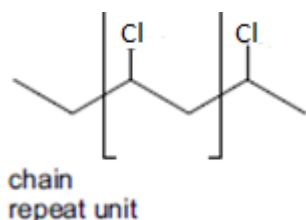
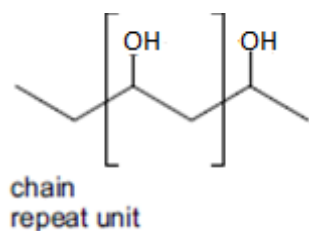
- LDPE has a less closely packed structure. It has less surface area of contact and weaker instantaneous dipole-induced dipole (id-id) attractions between polymer chains. Hence less energy is required to overcome the weak id-id attractions & melting point is lower.

- (b) Poly(vinyl alcohol), PVA, is a synthetic polymer used in papermaking, textiles, eye drops (such as artificial tears for dry eyes) and a variety of coatings. The monomer of PVA is vinyl alcohol, CH_2CHOH .

Poly(vinyl chloride), PVC, is a synthetic water-resistant polymer used for producing water pipes, textiles and clothings such as raincoats. The monomer of PVC is vinyl chloride, CH_2CHCl .

- (i) Draw a section of the PVA and PVC molecules containing at least 2 monomer molecules, and identify clearly the repeat units with [].

[2]



(allow displayed, structural or skeletal formulae)

- (ii) What type of polymerisation reaction is likely to have produced these polymers?

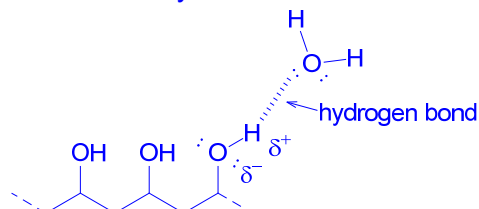
[1]

- Addition polymerisation

- (iii) With reference to the structure and bonding, explain why PVA is commonly used in eye drops while PVC is used in producing clothings such as raincoats.

[2]

- PVA is soluble in water. The **-OH group** in PVA forms hydrogen bonds between the chains and the water molecules. The water trap in the polymers is used to relieve eye dryness and soreness as it lubricates the surface of the eye.



- Raincoats made from PVC are water-resistant/repellent because PVC consist of id-id & pd-pd. It cannot absorb water.

- (iv) Suggest a reason why PVC might break down when exposed to concentrated aqueous sodium hydroxide whereas poly(ethene) does not.

- PVC reacts with NaOH, undergoes substitution reaction to form alcohol, bonds are disrupted and structure broken down.
- Poly(ethene) is non-polar and inert. It would not react and affect by NaOH.

[2]

[Total:10]

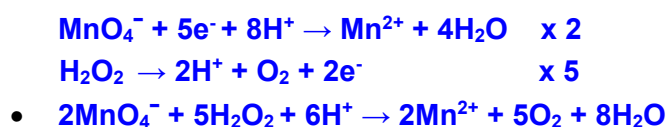
Section B

Answer **one** questions from this section, in the spaces provided

- 7 (a) Hydrogen peroxide is a mild antiseptic used on the skin to prevent infection of minor cuts, scarpes and burns.

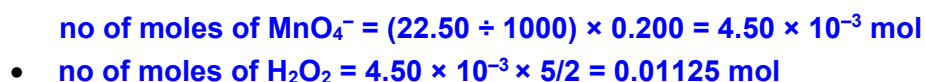
A stock solution of H_2O_2 was diluted by adding 40.0 cm^3 of the solution into a standard flask which was filled with distilled water to make a 250 cm^3 standard solution. 25.0 cm^3 of the standard solution was titrated with $0.200 \text{ mol dm}^{-3}$ of acidified KMnO_4 solution. 22.50 cm^3 of KMnO_4 solution was required to reach the end point and oxygen gas was produced at the end of reaction.

- (i) Write a balanced equation for the reaction between hydrogen peroxide and acidified KMnO_4 solution, showing your working clearly.



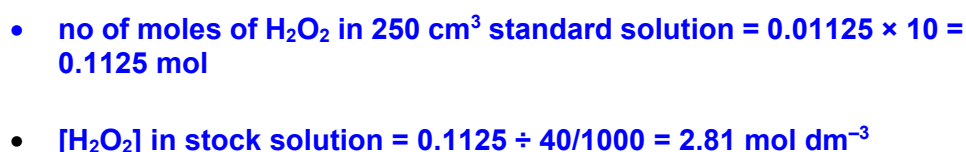
[1]

- (ii) Determine the amount of H_2O_2 , in moles, that reacted with KMnO_4 .



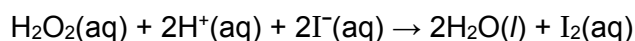
[1]

- (iii) Hence, determine the concentration of H_2O_2 in the stock solution.



[2]

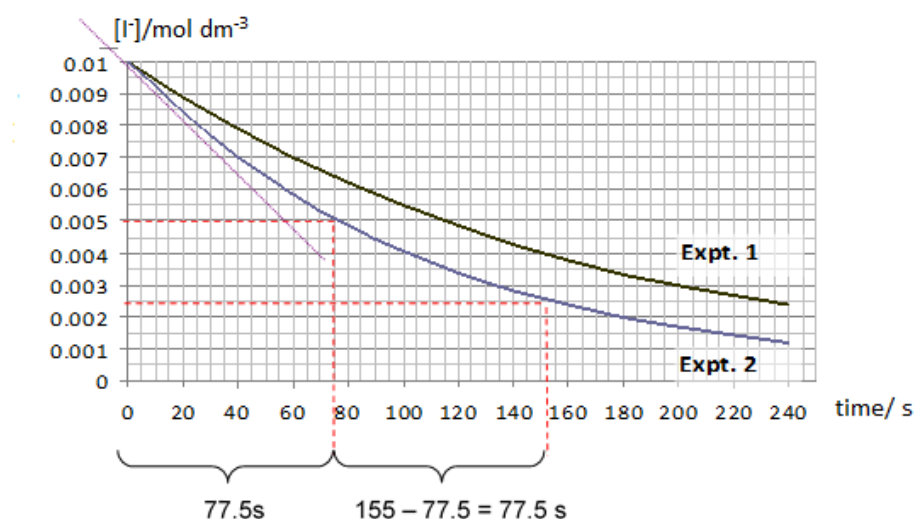
- (b) Hydrogen peroxide reacts with acidified iodide ions to liberate iodine according to the following reaction:



The rate for the reaction between iodide and aqueous H_2O_2 was studied using two solutions of different initial concentrations of H_2O_2 and the same initial concentration of iodide. The results are as shown below:

Time / s	Experiment 1	Experiment 2
	$[I^-] / \text{mol dm}^{-3}$ when $[\text{H}_2\text{O}_2] = 0.20 \text{ mol dm}^{-3}$	$[I^-] / \text{mol dm}^{-3}$ when $[\text{H}_2\text{O}_2] = 0.30 \text{ mol dm}^{-3}$
0	0.0100	0.0100
40	0.0079	0.0070
80	0.0062	0.0049
120	0.0049	0.0034
160	0.0038	0.0024
200	0.0030	0.0017

- (i) Plot a graph of these results, putting all the data on the same axes. Label each curve clearly.



[1m for correct label of axis and best fit smooth curves
1m for correct points plotted]

[2]

- (ii) From the graphs, deduce the order of reaction with respect to iodide and aqueous hydrogen peroxide.

- **constant $t_{1/2}$ of 77.5 s**
(construction lines for at least 2 half-lives must be shown on graph)
- **thus first order with respect to iodide.**

Expt 1: $0.20 \text{ mol dm}^{-3} \text{ H}_2\text{O}_2$,

$$\text{initial rate} = \frac{0.01 - 0.005}{85 - 0} = 5.88 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

Expt 2: $0.30 \text{ mol dm}^{-3} \text{ H}_2\text{O}_2$,

$$\text{initial rate} = \frac{0.01 - 0.004}{70 - 0} = 8.57 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

- When $[\text{H}_2\text{O}_2]$ increases 1.5 x, initial rate increases $\frac{8.57 \times 10^{-5}}{5.88 \times 10^{-5}} = 1.46 \text{ x}$
- 1st order w.r.t. H_2O_2 .

[4]

(iii) The order of reaction with respect to H^+ is zero. Hence, write the rate equation for the reaction.

- rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$

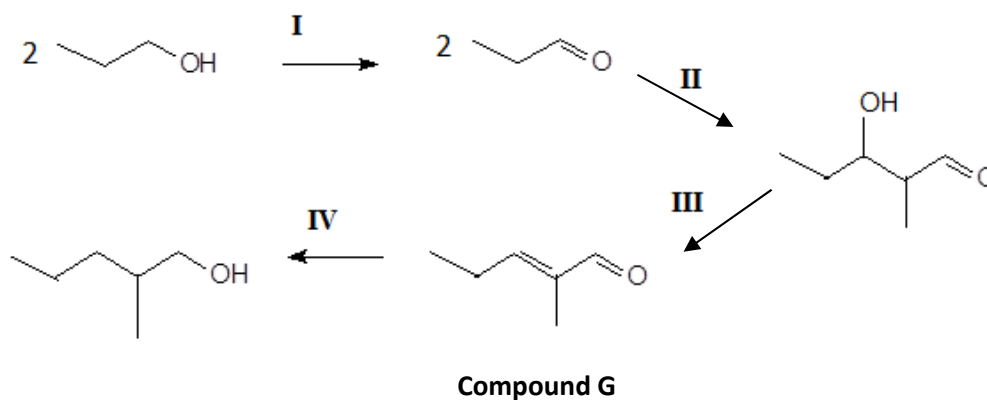
[1]

(iv) Calculate the rate constant for **experiment 2**, stating its units.

- $k = \frac{8.57 \times 10^{-5}}{(0.30)(0.01)} = 0.0286 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$
- or $k = \frac{\ln 2}{77.5 \times 0.3} = 0.0298 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$

[1]

(c) A four-step process to convert propanol into 2-methylpentan-1-ol could be an important step for production of renewable energy.



(i) State the reagents and conditions required for steps I and IV.

- **step I:** $\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 , warm with immediate distillation
- **step IV:** H_2 , Pt / Pd, room temperature (accept Ni, with elevated temp and pressure)

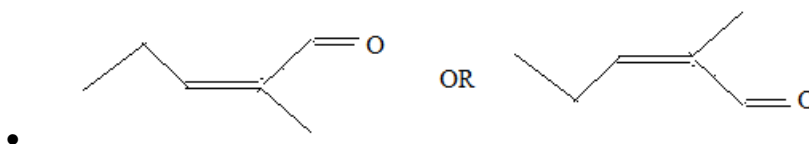
[2]

(ii) By considering the change in molecular formulae shown in steps II and III, suggest the types of reaction that occurred in both steps.

- **step II:** **addition**
- **Step III:** **elimination**

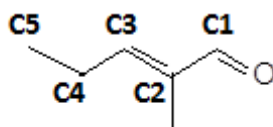
[2]

(iii) Draw the skeletal formula of the other stereoisomer of compound G.



[1]

(iv) By considering the hybridization of the carbon atoms labelled C1, C2, C4 and C5 in compound G below, predict which sigma bond, C1-C2 or C4-C5, will be shorter and explain your answer.

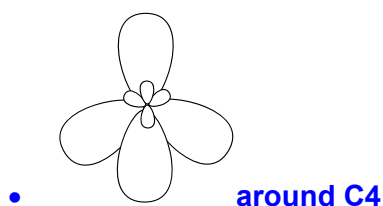


Compound G

- **The C1-C2 bond is shorter**
- **C1-C2 is $\text{sp}^2\text{-sp}^2$ orbital overlap, while C4-C5 is $\text{sp}^3\text{-sp}^3$ orbital overlap. sp^2 orbital has more s character, resulting in more effective orbital overlap and stronger bond.**

[2]

- (v) Draw the hybrid orbitals around the carbon atom labelled **C4**.



[1]

[Total: 20]

- 8 (a) Describe the thermal decomposition of the hydrogen halides HCl, HBr and HI and explain any variation in their thermal stabilities.

- Hydrogen halides decompose thermally to form H_2 gas and halogen, X_2 .
 - Thermal stability decreases down the group: $HCl > HBr > HI$
 - This is because the atomic radius increases down the group, resulting in decreasing degree of effective overlap of orbitals and covalent bond strength and hence bond energy of HX decreases down the group.
- (alternative ans: quote bond energy values from Data Booklet and recognise that H-X bond strength decreases in the order $HCl > HBr > HI$)

[3]

- (b) Halogenoalkanes can be made by the reaction of the corresponding halogen with alkanes.

When propane is treated with a small quantity of chlorine in the presence of ultraviolet light, two mono-substituted products are produced.

The order of reactivity of tertiary, secondary and primary hydrogen atoms follows a 5:4:1 ratio.

Draw the structural formula of the two mono-substituted products, and state the approximate ratio in which they are formed.

- $CH_3CH_2CH_2Cl$: $CH_3CHClCH_3$

6 x 1	:	2 x 4
6	:	8
3	:	4

[2]

- (c) Compound **A** ($M_r=116$) is a sweet smelling substance. Acid hydrolysis of compound **A** yields two products, compounds **B** and **C**. **C** reacts with Na_2CO_3 to give CO_2 .

A series of tests were performed on compound **B**.

- (i) **B** has the molecular formula $\text{C}_4\text{H}_{10}\text{O}$. Reacting **B** with acidified potassium manganate(VII) decolourises the solution and produces compound **D**, $\text{C}_4\text{H}_8\text{O}_2$. Heating **B** with excess concentrated sulfuric acid produces compound **E**, which does not show cis-trans isomerism. Reaction of **E** with hydrogen gas forms 2-methylpropane.

Suggest structures for **B**, **D** and **E** and explain the reactions involved.

•**B**

Compound	•••Structure
E	$\begin{array}{c} \text{CH}_3-\text{C}=\text{CH}_2 \\ \\ \text{CH}_3 \end{array}$
B	$\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{OH} \\ \\ \text{CH}_3 \end{array}$
D	$\begin{array}{c} \text{CH}_3-\text{CH}-\text{COOH} \\ \\ \text{CH}_3 \end{array}$

undergoes oxidation with acidified KMnO_4 to form **D**.

→ From **B** to **D**, there is an increase in 1 oxygen atom per molecule

→ **B** is a primary alcohol (contains $-\text{CH}_2\text{OH}$ group) and **D** is a carboxylic acid.

•**B** undergoes elimination when heated with excess conc. sulfuric to form **E**.

→ presence of $\text{C}=\text{C}$ bond in **E**

[5]

- (ii) Hence, suggest a possible structure for compound **A**.

$$M_r \text{ of C} = M_r \text{ of A} - M_r \text{ of B} + M_r \text{ of H}_2\text{O}$$

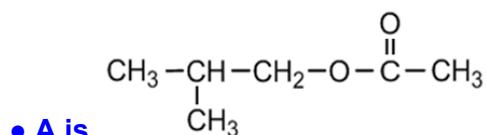
$$= 116 - [4(12)+10(1)+16] + 18$$

$$= 60$$

•**C** undergoes acid-carbonate reaction with Na_2CO_3 to give CO_2

→presence of $-\text{COOH}$ group in **C**

→structure of **C** is most likely CH_3COOH



[2]

- (d) A student wishes to investigate the kinetics of the acid-catalysed hydrolysis of compound **A** described in (c).
- (i) A number of experiments at a constant pressure of 101 kPa and temperature of 90 °C were performed in which the concentrations of compound **A** and hydrogen ions were varied. The results are shown below:

Experiment	[H ⁺] / moldm ⁻³	[compound A] / moldm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.0100	0.120	1.8 × 10 ⁻⁴
2	0.0050	0.240	1.8 × 10 ⁻⁴
3	0.0050	0.480	3.6 × 10 ⁻⁴

Using the above data, deduce the order of reaction with respect to compound **A** and hydrogen ions.

From experiments 2 and 3,

• When [A] doubles and [H⁺] remains constant, the initial rate doubles. Hence, the reaction is first order with respect to A.

• Let the order of reaction with respect to H⁺ be x.

$$\frac{\text{rate 1}}{\text{rate 2}} = \frac{[\text{H}^+]^x[\text{A}]}{[\text{H}^+]^x[\text{A}]}$$

$$\frac{1.8 \times 10^{-4}}{1.8 \times 10^{-4}} = \frac{[0.0100]^x[0.120]}{[0.0050]^x[0.240]}$$

$$x = 1$$

Hence, the reaction is first order with respect to H⁺.

[2]

- (ii) Hence, construct a rate equation and use it to calculate a value of the rate constant, k, including its units, for the acid-catalysed hydrolysis of compound **A**.

Rate = k [H⁺][A]

$$\bullet 1.8 \times 10^{-4} = k(0.0100)(0.120)$$

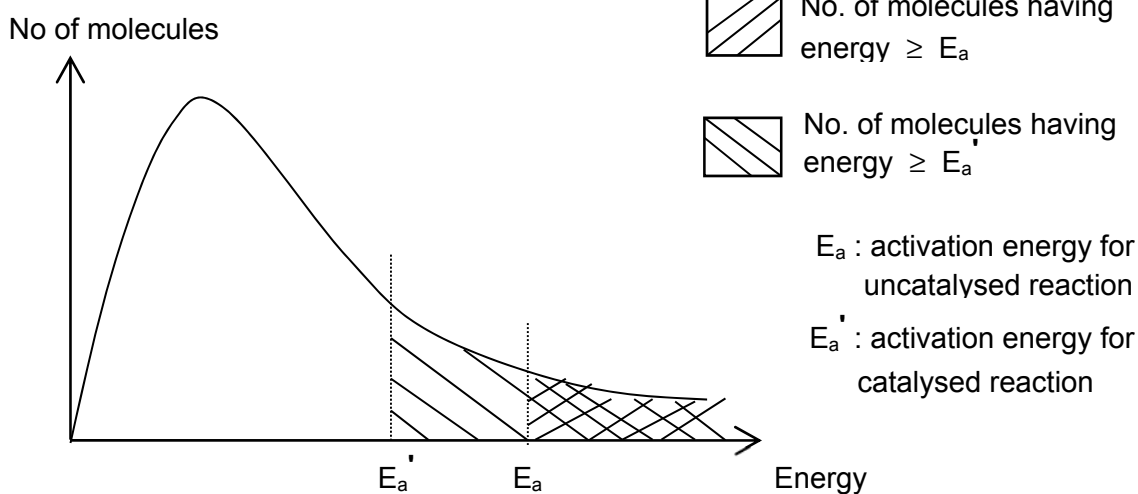
$$\bullet k = 0.15 \text{ mol}^{-1}\text{dm}^3\text{s}^{-1} \text{ (1m working, 1m correct ans \& units)}$$

[2]

(iii) With reference to a diagram where necessary, state and explain how the rate of reaction would change when the experiment was conducted:

- in the absence of hydrogen ions
- at a pressure of 121 kPa

In the absence of H^+



• diagram

• A catalyst increases the rate of reaction by providing an alternative pathway which involves a lower activation energy than that of an uncatalysed reaction.

• In the absence of the catalyst H^+ , the number of reactant molecules having energy greater than or equal to the lower activation energy, E_a decreases significantly. Hence the frequency of effective collisions decreases and the rate decreases.

At a pressure of 121 kPa

• When the pressure of the reaction increases (temperature remaining the same), the rate of reaction remains the same as the reactants are in the aqueous state and are not affected by pressure.

[4]

[Total: 20]